

03_CANONICAL_DEFINITIONS_AND_TAXONOMY .md

Canonical Definitions, Ecosystem Nomenclature, System Mottoes & Architectural Classifications

Scope: Authoritative consolidation of all definition documents, brand nomenclatures, mottos, and structural taxonomies from source folder **7-DEFINITIONS-(4-files)**.

1. The Core System Identity & Elevator Pitch

The entire ecosystem is formally defined as:

"A Local Agentic Cognitive Architecture backed by an Open Memory Protocol (OB1) and a Structured Markdown/JSONL Knowledge Pipeline."

Technically categorized as:

- **Federated, Multi-Tier Cognitive Architecture with an Active Agentic Knowledge Layer**
- **Hybrid Cognitive Memory Engine (CME)**
- **Cross-Platform Context Router & Graph Knowledge Graph (GKG)**

Unlike traditional static RAG, this architecture splits cognitive processing into immediate conversational context, graph-based associative reasoning, structured semantic documentation, and permanent relational storage.

2. Ecosystem Mottoes & Philosophical Foundations

1. Family Motto / Ecosystem Axiom:

- Latin: *Xçineribus, in-variis-nunquam-varius, Novi-Æxentis-Copiae, Vincent*
- Translation: *"From the ashes, unwavering in the midst of adversity, new forces of abundance conquer."*
- Symbolism: The Ash tree (Yggdrasil)—the hidden, resilient root network growing quietly out of the ashes, connecting disparate computational nodes into an unwavering living body.

2. Universal Memory Standard (#d.u.m.b.a.s.s.):
 - "It isn't 'dumbass-proof' if it hasn't been '#d.u.m.b.a.s.s.' proven."™
 - Meaning: No tool, agent, or workflow is certified production-ready until it has passed zero-loss recovery and integrity verification under the #d.u.m.b.a.s.s. state engine.
3. Frontend Concierge Slogan:
 - "~Ever-evolving ecosystems, for ever-expanding Horizons."™
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3. Canonical Subsystem Taxonomy & Brand Nomenclature

The architecture decomposes into five sovereign tiers and decoupled on-device execution daemons:

System Component	Git Repository Name	UI Branding	Architectural Role & Symbolism
Presentation Shell	horizons-ui/	Horizons UI™	The visible canopy & concierge. Android Chromium WebView running HTML5/React tiles with persistent WebSocket bridges to backend daemons.
Cognitive Brain	novus-aexenti/	NovusÆxenti™	The dynamic agent core. Asymmetric dual-brain MoE coordinating 0.8B fast intent triage and 4B/9B deep code execution.
Action Surface	novaecopia/	NovÆcopia™	The abundance of the system. Hosts native Android execution daemons (Æsc & Æyre), OpenWiki TUI, modular harnesses,

System Component	Git Repository Name	UI Branding	Architectural Role & Symbolism
			and MCP server bridges.
Orchestration Layer	aesop-xi/	Æsop-Xi™	The governing infrastructure. Enforces inter-agent resource arbitration, security clearance, tool guardrails, and isolated Red Auditor gatekeeping.
Universal Memory	novae-xorpus/	NovÆxorpus™	The hardened trunk. Universal database, cold sensory archive (Tier 1), living LLM wiki (Tier 2), and machine recall cache (Tier 3).

Decoupled On-Device Daemons:

- **Æsc (aesc APK):** Native System Terminal & Shell Daemon. Replaces Termux by registering as an OS Accessibility Service/Assistant. Hosts local ADB loopback on [127.0.0.1:5555](#) to execute shell commands with UID 2000 privileges without root.
- **Æyre (aeyre APK):** Sensory Ingress & Media Daemon. Runs bare-metal Silero VAD audio stream, Moonshine Small ONNX speech-to-text, Kokoro-82M text-to-speech, and Video Game SDK screen capture.

4. Hybrid Cognitive Memory Engine (CME) Breakdown

The memory architecture organizes across four distinct temporal and functional states:

1. **In-Context Short-Term Layer (mem0 + OmniRoute):**
 - **mem0:** Manages in-session rolling habits, dynamic user syntax preferences, and short-term personalization.
 - **OmniRoute** (<http://localhost:20128/v1>): Serves as both the unified data routing gateway (directing requests across NPU, Jetson, and Cloud) and an

asynchronous memory tap extracting candidate reasoning paths into the Reasoning Bank.

2. **Structural Medium-Term Layer ([wiki_md/](#)):**

- Implements Andrej Karpathy's Living LLM Wiki standard via OpenWiki TUI. A compounding, human- and machine-readable Markdown second brain with bidirectional [\[\[wikilinks\]\]](#).

3. **Global Persistence Layer ([OB1 / Open Brain](#)):**

- Nate B. Jones's OB1 protocol operating over PostgreSQL with [pgvector](#) via MCP. Decouples memory from inference ("own the memory, rent the model") so any model instantly inherits system knowledge.

4. **Cognitive Execution History ([reasoning_bank/](#)):**

- The system flywheel and epistemic ledger ([active_execution_paths.json](#)). Stores *how* problems were solved, step-by-step logic trajectories, and RLVR validation outcomes (+1.0 / -1.0) to ensure zero-loss crash recovery.

5. Master Orchestration & Execution Router

- **Orchestrator Role:** The central workflow router directing tasks based on complexity:
 - Low-latency, bounded on-device tasks → Local 0.8B/9B weights running on Snapdragon Hexagon NPU.
 - Intermediate compilation and local indexing → Node Beta (NVIDIA Jetson Orin Nano Super).
 - High-compute frontier analysis → Cloud providers via OpenRouter / Vertex AI.
- **Tooling Integration:** Integrates Everything Claude Code (ECC), Honey-for-Devs, AST code dependency mappers (Graphify), and Matt Pocock prompt skills into modular, repeatable execution environments.